

Answer **all** the questions in the spaces provided.

- 1 (a) Define *velocity*.

Rate of change of displacement

[1]

- (b) The speed v of a sound wave through a gas of pressure P and density ρ is given by the equation

$$v = \sqrt{\frac{kP}{\rho}}$$

where k is a constant that has no units.

An experiment is performed to determine the value of k . The data from the experiment are shown in Fig. 1.1.

quantity	value	uncertainty
v	$3.3 \times 10^2 \text{ m s}^{-1}$	$\pm 3\%$
P	$9.9 \times 10^4 \text{ Pa}$	$\pm 2\%$
ρ	1.29 kg m^{-3}	$\pm 4\%$

Fig. 1.1

- (i) Use data from Fig. 1.1 to calculate k .

$$\frac{(3.3 \times 10^2)^2 \times 1.29}{9.9 \times 10^4} = 1.419$$

$$k = 1.4 \quad [2]$$

- (ii) Use your answer in (b)(i) and data from Fig. 1.1 to determine the value of k , with its absolute uncertainty, to an appropriate number of significant figures.

$$k = \frac{v^2 \times \rho}{P}$$

$$\% \text{ Uncertainty} = (3 \times 2) + 2 + 4 = 12 \%$$

$$\text{Absolute Uncertainty} = 1.419 \times \frac{12}{100} = 0.17028 \approx 0.2$$

$$k = 1.4 \pm 0.2 \quad [3]$$

[Total: 6]

- 3 The variation with extension x of the force F acting on a spring is shown in Fig. 3.1.

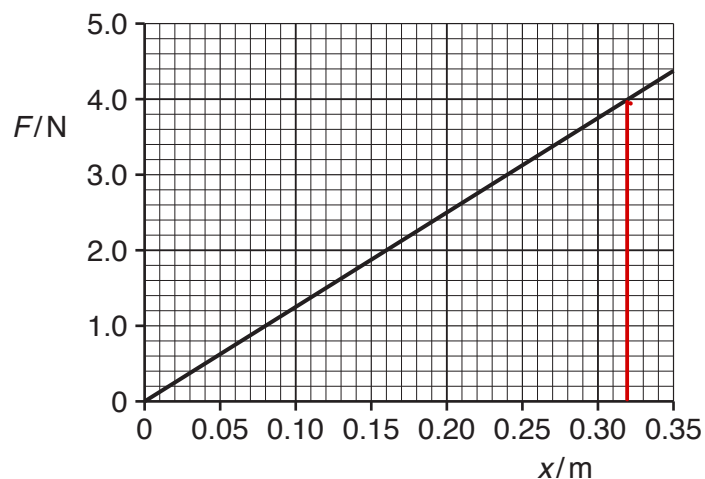


Fig. 3.1

The spring of unstretched length 0.40 m has one end attached to a fixed point, as shown in Fig. 3.2.

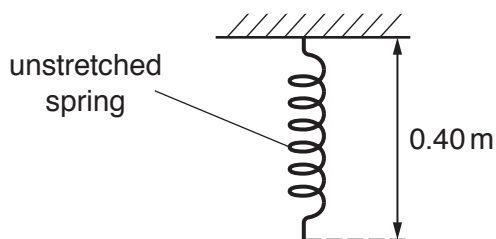


Fig. 3.2

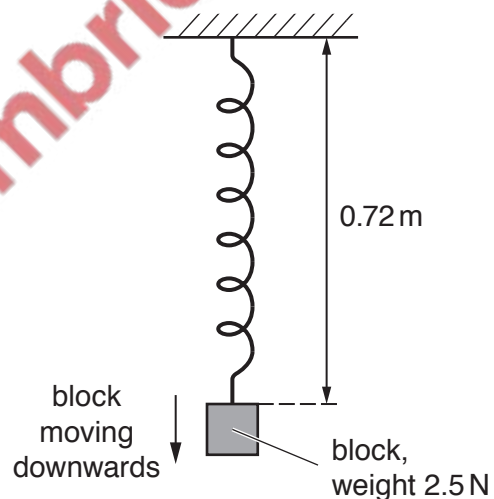


Fig. 3.3

A block of weight 2.5 N is then attached to the spring. The block is then released and begins to move downwards. At one instant, as the block is continuing to move downwards, the spring has a length of 0.72 m, as shown in Fig. 3.3.

Assume that the air resistance and the mass of the spring are both negligible.

(a) For the change in length of the spring from 0.40 m to 0.72 m:

(i) use Fig. 3.1 to show that the increase in elastic potential energy of the spring is 0.64 J

$$\begin{aligned}
 x &= 0.72 - 0.40 = 0.32 \\
 E &= \text{area under graph} \\
 &= \frac{1}{2} \times 0.32 \times 4 \\
 &= 0.64 \text{ J}
 \end{aligned}$$

[2]

(ii) calculate the decrease in gravitational potential energy of the block of weight 2.5 N.

$$\begin{aligned}
 \Delta GPE &= mg \Delta h \\
 &= \frac{2.5}{9.81} \times 9.81 \times 0.32 \\
 &= 2.5 \times 0.32 \\
 &= 0.8 \text{ J}
 \end{aligned}$$

decrease in potential energy = 0.80 J [2]

(b) Use the information in (a)(i) and your answer in (a)(ii) to determine, for the instant when the length of the spring is 0.72 m:

(i) the kinetic energy of the block

↑ EPE
↓ GPE

$$\begin{aligned}
 K_E &= 0.80 - 0.64 \\
 &= 0.16
 \end{aligned}$$

kinetic energy = 0.16 J [1]

(ii) the speed of the block.

$$\begin{aligned}
 0.16 &= \frac{1}{2} \times \frac{2.5}{9.81} \times v^2 \\
 v &= 1.12 \text{ ms}^{-1} \\
 &\approx 1.1 \text{ ms}^{-1} \text{ (2sf)}
 \end{aligned}$$

speed = 1.1 ms⁻¹ [2]

[Total: 7]